

Data Flow Diagram

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram:

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	User / Farmer / Researcher: Sends soil & environmental arrays to the platform; receives dynamic crop diagnostics.
Rectangle with numbered header	Process	1 Input Validation & Parsing: Intercepts and validates numeric ranges for chemical data (N, P, K, pH) and weather vectors. 2 ML Inference Execution: Uses the serialized Random Forest model array (model.pkl) to execute classification rules.
Rectangle (solid fill, no number)	Data Store	Model Store (model.pkl): Persisted classification weights generated during training. Crop Insights Matrix: Context dictionary matching crop targets to agro-ecological zone constraints.
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example:

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

